Introduction to Machine Learning

A Simple Study Guide for RAG Assignment Testing

1. What is Machine Learning?

Machine Learning (ML) is a branch of artificial intelligence that enables computers to learn patterns from data and make predictions or decisions without being explicitly programmed for every situation. A machine learning system usually learns from examples, evaluates its performance, and improves when it receives useful training data.

For example, an email application can learn from previously labelled messages to identify whether a new email is likely to be spam. The quality of the result depends on the quality and relevance of the training data.

2. Types of Machine Learning

There are three commonly discussed types of machine learning: supervised learning, unsupervised learning, and reinforcement learning.

Supervised Learning

Supervised learning uses labelled examples. Each training example contains input data and a known target or answer. The model learns a relationship between the input and the target. It is used for tasks such as predicting house prices and classifying emails as spam or not spam.

Unsupervised Learning

Unsupervised learning works with data that does not have labelled answers. The model looks for structure or patterns, such as grouping customers with similar purchasing behaviour. Clustering is a common unsupervised learning task.

Reinforcement Learning

Reinforcement learning involves an agent that learns by taking actions and receiving rewards or penalties from its environment. Over time, the agent learns a strategy that aims to increase its total reward. It can be applied to robot control and game-playing systems.

3. Basic Machine Learning Workflow

1. Collect data relevant to the problem.
2. Clean the data and handle missing or incorrect values.
3. Split the data into training and testing sets.
4. Choose an appropriate machine learning algorithm.
5. Train the model using the training data.
6. Evaluate the model on testing data using suitable metrics.
7. Use the trained model to make predictions on new data.

4. Common Applications

- Healthcare: supporting analysis of medical data.
- Education: identifying learning patterns and supporting personalized practice.
- Finance: detecting unusual transactions and estimating risk.
- Transportation: predicting traffic patterns and supporting route planning.
- Retail: recommending products based on customer preferences.

5. Advantages and Limitations

Advantages: Machine learning can discover patterns in large datasets, automate some predictions, and support data-informed decisions. Limitations: Results may be inaccurate when data is incomplete, biased, outdated, or different from the data used during training. Models also need suitable evaluation and monitoring.

6. Quick Revision Questions

1. What is machine learning?
2. What are the three commonly discussed types of machine learning?
3. How does supervised learning differ from unsupervised learning?
4. What is the role of rewards in reinforcement learning?
5. Why do we split data into training and testing sets?
6. Give two real-world applications of machine learning.